

TechCorp Enterprise

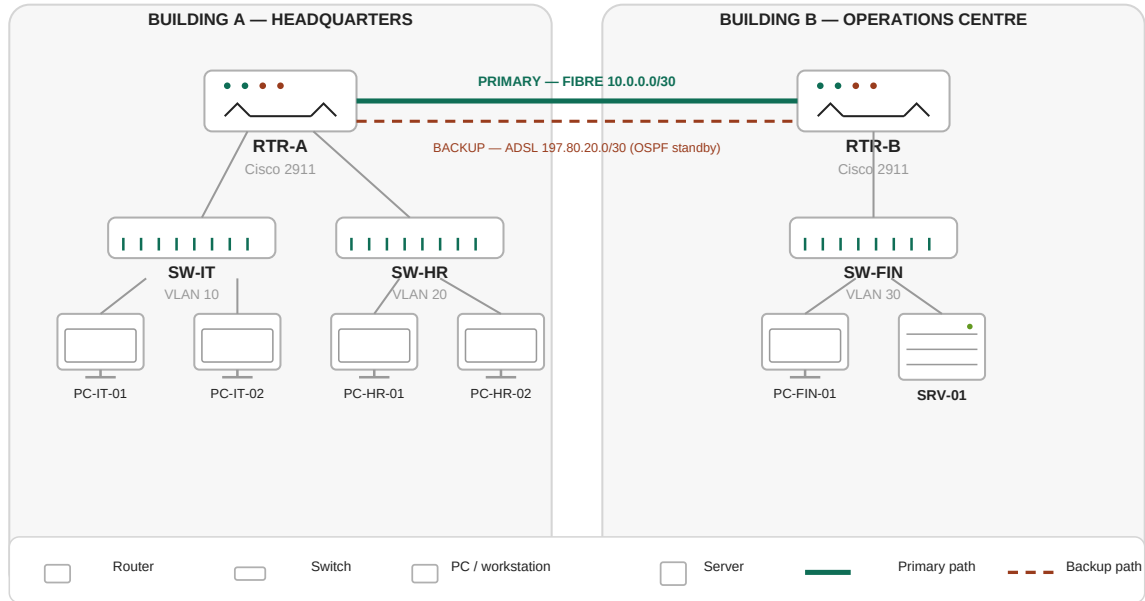
Resilient WAN Failover Architecture

Dual-WAN Redundancy, Floating Static Routes, OSPF-Based Automatic Failover & Recovery Validation — Cisco Packet Tracer

Dual WAN Links	Floating Static Routes	OSPF Failover	Failover Testing
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Document Reference	TC-NID-2025-002 — Portfolio Edition
Candidate	Tiisetso Tlotlo Hlongwane
Environment	Cisco Packet Tracer
Scope	Backup WAN Provisioning, Routing & Automated Failover (Phase 1–5)
Date	2026

The diagram below summarises the dual-WAN topology this project builds: two departmental networks in Building A and the Finance network in Building B, connected across two independent WAN circuits between RTR-A and RTR-B. Under normal conditions all inter-site traffic flows over the primary fibre path; the ADSL path remains on standby and is brought into service automatically if the primary path fails.



Why it matters: Laying out the topology visually before the configuration detail gives a reader the full picture in seconds — which paths exist, which is preferred, and which department sits behind which switch — context that is otherwise only reconstructed by reading every phase in sequence.

Objective

This portfolio documents the design, configuration, and validation of a resilient dual-WAN architecture for TechCorp, built and tested inside Cisco Packet Tracer as a direct continuation of the Phase 1 network. TechCorp's inter-office connection relied on a single fibre WAN circuit between Building A and Building B — a single point of failure that would sever all inter-site connectivity if the link were cut or the provider suffered an outage. This project introduces a second, independently addressed WAN path over ADSL and configures both routers so that traffic automatically reroutes to the backup path within seconds of a primary failure, and automatically reverts once the primary path recovers — with zero manual intervention at any point.

The objective throughout was to demonstrate not just that each configuration step could be executed, but why each design decision was made, and to prove — through deliberate, controlled failure testing — that the resulting network behaves as an enterprise redundancy design should.

Scope & Context

Building on the existing Phase 1 topology (two Cisco 2911 routers, RTR-A and RTR-B, connected over a primary fibre WAN circuit) this phase adds a second serial WAN interface to each router and a dedicated ADSL backup path between the two buildings.

New WAN Addressing Introduced

Device	Interface	IP Address	Role
RTR-A	Se0/0/0	10.0.0.1 /30	Primary WAN (fibre) — from Phase 1
RTR-A	Se0/0/1	197.80.20.1 /30	Backup WAN (ADSL) — new
RTR-B	Se0/0/0	10.0.0.2 /30	Primary WAN (fibre) — from Phase 1
RTR-B	Se0/0/1	197.80.20.2 /30	Backup WAN (ADSL) — new

Deliverables Covered in this Portfolio

Area	Evidence Captured
Backup WAN Provisioning	Second serial interface added and cabled on both routers as an independent ADSL path
Floating Static Routes	AD-based primary/backup routes configured and verified as hidden until needed
Automated Failover	OSPF extended across both WAN links to provide self-healing failure detection
Failover Validation	Controlled link failure and recovery testing with before/during/after evidence

Outcome

All planned configuration was implemented and verified with direct evidence captured at every stage: a second physical WAN path provisioned and addressed; floating static routes proven to remain hidden under normal conditions; an OSPF-based failover mechanism that reroutes traffic automatically on primary link failure; and a full failure-and-recovery test cycle confirming the network heals itself without administrator action. The following

sections present this work as a structured, phase-by-phase evidence log.

This section explains a deliberate mid-project design change: the plan for automated failover was originally IP SLA with a tracked object bound to the floating static route. During implementation this could not be fully realised in Packet Tracer, so the design was revised to use OSPF instead — itself a valid, widely deployed enterprise redundancy pattern.

FIGURE 2.1

Why the Original Plan Changed

The 2911 router model in Cisco Packet Tracer does not fully support binding a track object to a static route in the way real Cisco IOS does. Rather than leave the failover mechanism half-built, the design was revised to extend OSPF across both WAN interfaces, which is fully supported in the simulator and achieves the same self-healing outcome.

Why it matters: A design that adapts to a platform constraint and still meets the original requirement is more valuable evidence than one that simply stops at the first obstacle — this reflects how real network engineering decisions get made.

FIGURE 2.2

How OSPF-Based Failover Works

- **Normal operation:** both WAN interfaces are advertised into OSPF; the lower-cost fibre link is preferred and carries all inter-site traffic while the ADSL link remains known but unused.
- **On failure:** OSPF Hello packets are exchanged every 10 seconds; if a neighbour is not heard from within the 40-second dead interval the adjacency is declared down, routes through that interface are withdrawn, and the shortest path is recalculated so the ADSL link is installed automatically.
- **On recovery:** OSPF re-establishes the adjacency once the fibre link returns and automatically prefers it again due to its lower cost.

FIGURE 2.3

IP SLA vs OSPF — Trade-off Comparison

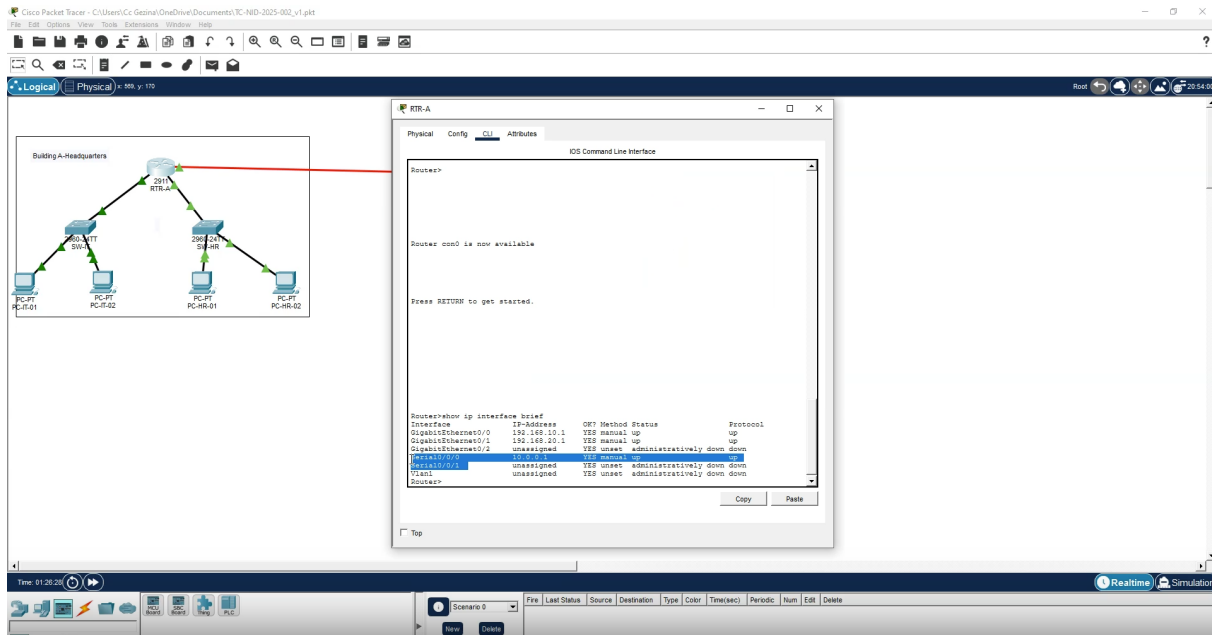
Feature	IP SLA + Track (planned)	OSPF (implemented)
Failover trigger	Active probe — detects failure even if interface stays up	Neighbour loss via Hello/dead timer
Failover time	Under 10 seconds	Up to 40 seconds (dead interval)
Packet Tracer support	Partial	Full
Real-world use	Preferred where sub-10s convergence is required	Common, widely deployed

Why it matters: The 40-second convergence time is the only real trade-off against the original plan. For most enterprise use cases this window is acceptable, and OSPF-based redundancy is a legitimate, standard design pattern rather than a compromise.

The existing HWIC-2T module on both routers was confirmed to provide a second serial interface (Se0/0/1). A second serial DCE cable was connected between RTR-A Se0/0/1 and RTR-B Se0/0/1, establishing the physical path that the ADSL backup link would run over, alongside the existing Se0/0/0 fibre path.

FIGURE 3.1

Second Serial Port Confirmed

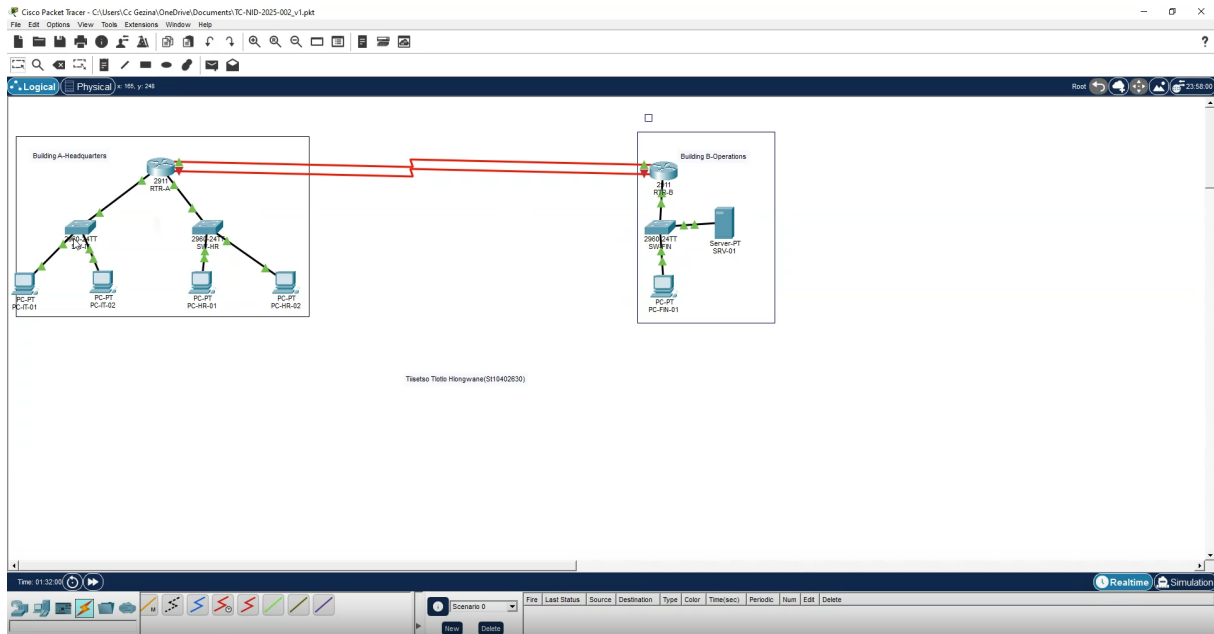


Checked show ip interface brief on both routers before making any changes, confirming the HWIC-2T module already exposed a second serial interface from the original Phase 1 build.

Why it matters: Verifying hardware availability before configuring anything avoids wasted troubleshooting later — if the port were missing, the fault would need to be resolved at the physical layer first, not chased through IP or routing configuration.

FIGURE 3.2

Backup Serial Cable Connected



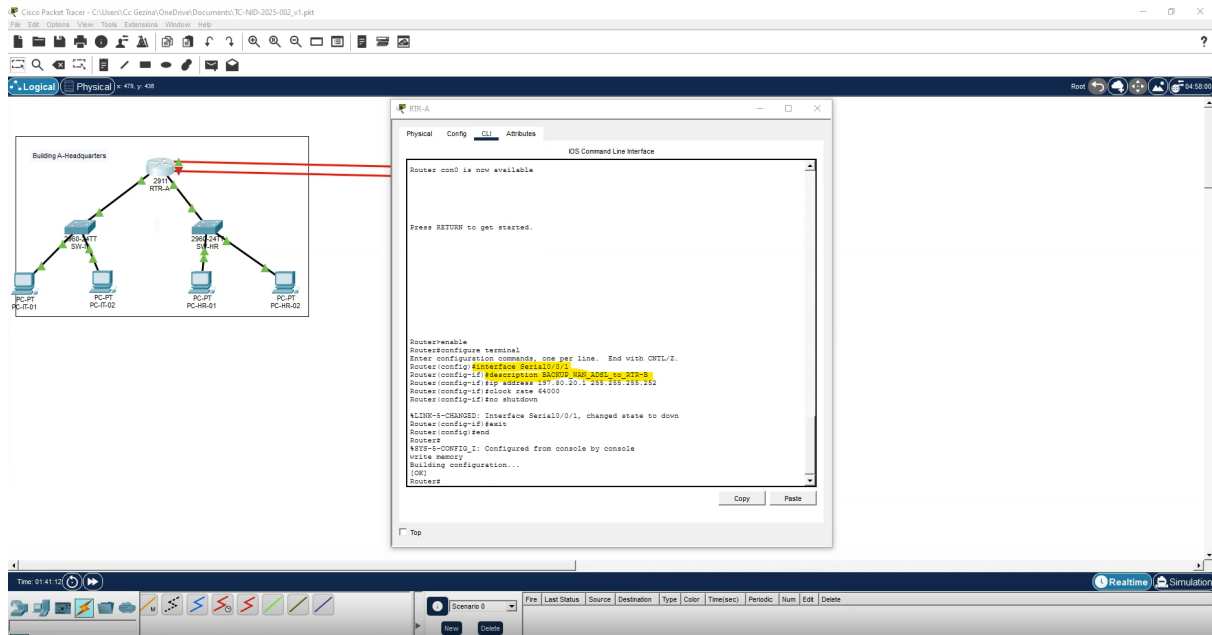
Added a second Serial DCE cable between RTR-A Se0/0/1 and RTR-B Se0/0/1, creating a fully independent physical path between the two buildings, separate from the existing fibre link.

Why it matters: Using a distinct cable and interface pair for the backup path means a physical fault on the primary link — a cut cable or damaged port — cannot also take down the backup path.

IP addresses were assigned to the new Se0/0/1 interfaces on both routers, the interfaces were enabled, and clocking was configured on the DCE end (RTR-A). The link was verified as up/up on both ends, and connectivity was confirmed with a successful ping across the new backup path.

FIGURE 4.1

Backup Interface Addressed

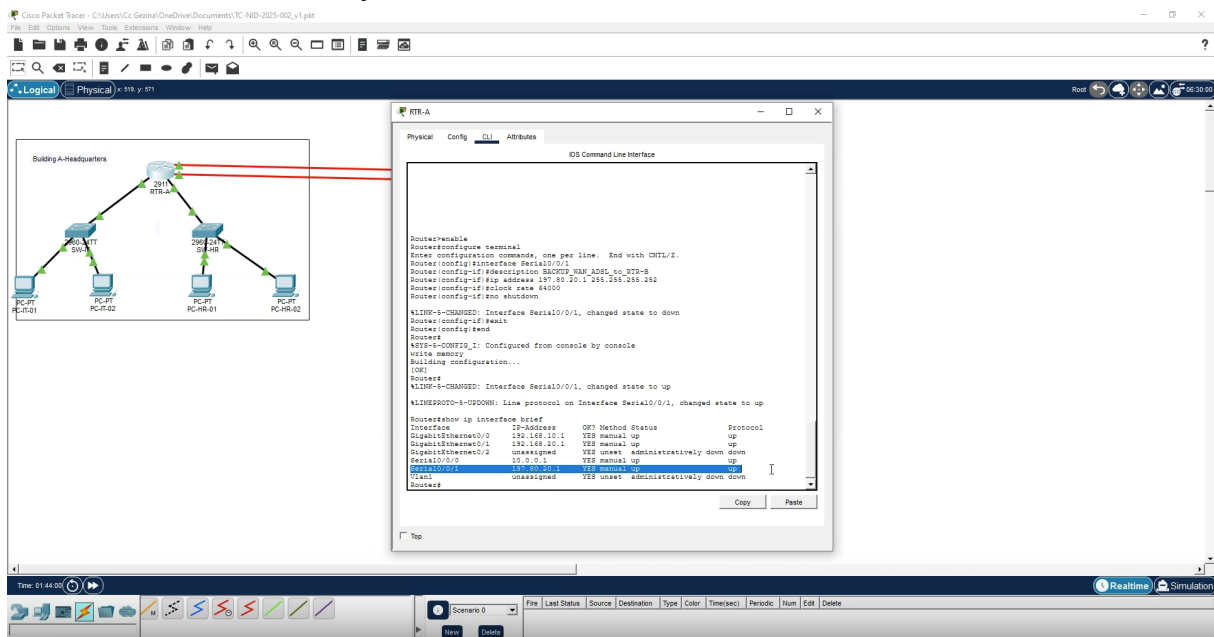


Assigned 197.80.20.1/30 and 197.80.20.2/30 to Se0/0/1 on RTR-A and RTR-B respectively, and set the clock rate on RTR-A as the DCE end of the link.

Why it matters: The DCE end of a serial link supplies clocking for the connection; without it the link stays physically connected but never comes up at Layer 2 — a fault that would otherwise look identical to a cabling problem.

FIGURE 4.2

Both Serial Links Verified Up

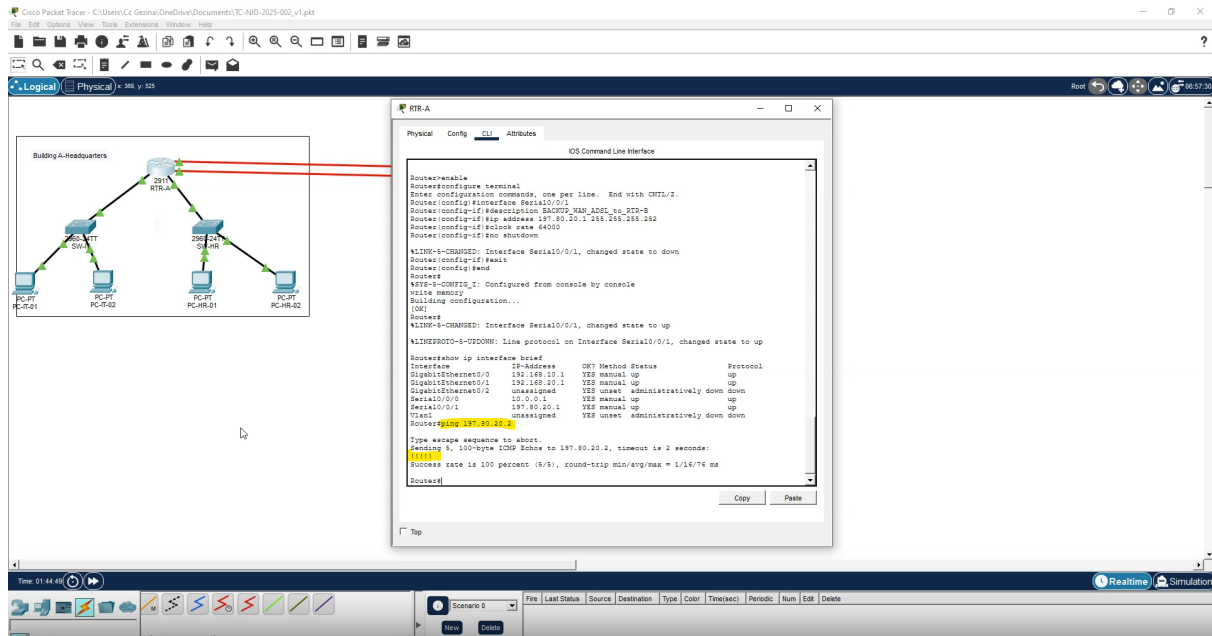


Ran show ip interface brief on RTR-A and confirmed both Se0/0/0 (primary) and Se0/0/1 (backup) report up/up status.

Why it matters: Confirming Layer 1/2 status before testing Layer 3 reachability isolates where a fault would sit if the following ping test failed.

FIGURE 4.3

Backup Path Connectivity Confirmed



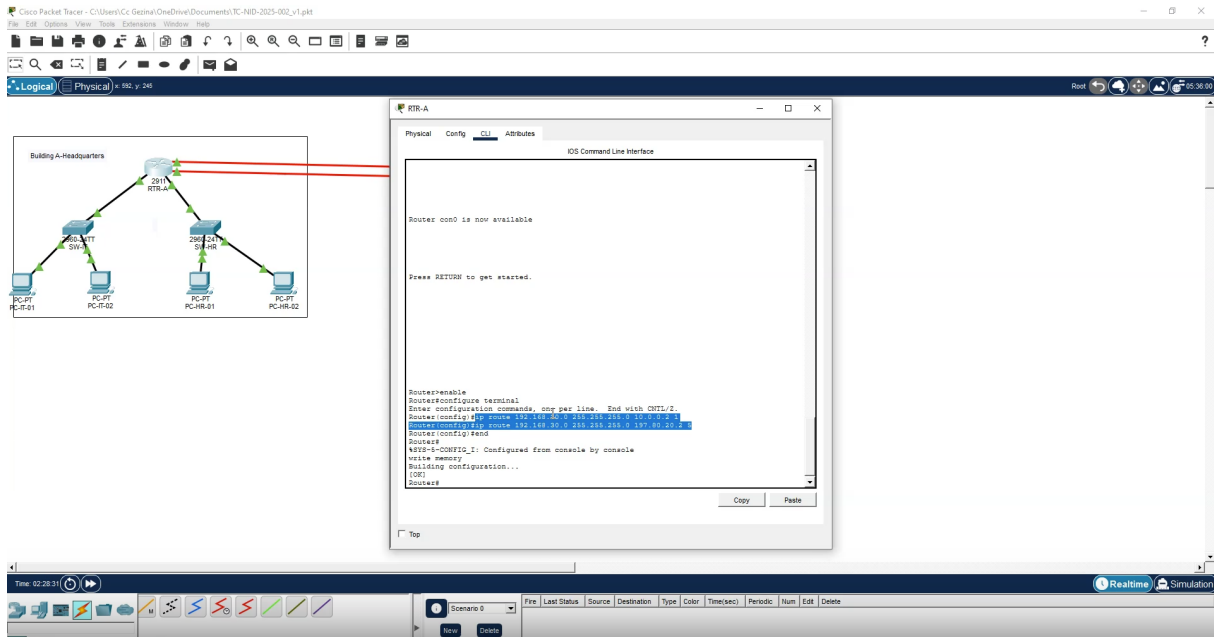
Pinged 197.80.20.2 from RTR-A and received a full 5/5 success rate, proving the backup ADSL path is functional end-to-end at the network layer.

Why it matters: This is the checkpoint that proves the backup path is a genuinely usable alternative route, not just a cable that happens to be connected.

Floating static routes were configured on both routers for every relevant subnet: a primary route with Administrative Distance 1 via the fibre link, and a backup route with Administrative Distance 5 via the ADSL link. Because the router always installs the route with the lowest AD, the backup route remains hidden in the routing table until the primary route is withdrawn.

FIGURE 5.1

Primary and Backup Routes on RTR-A

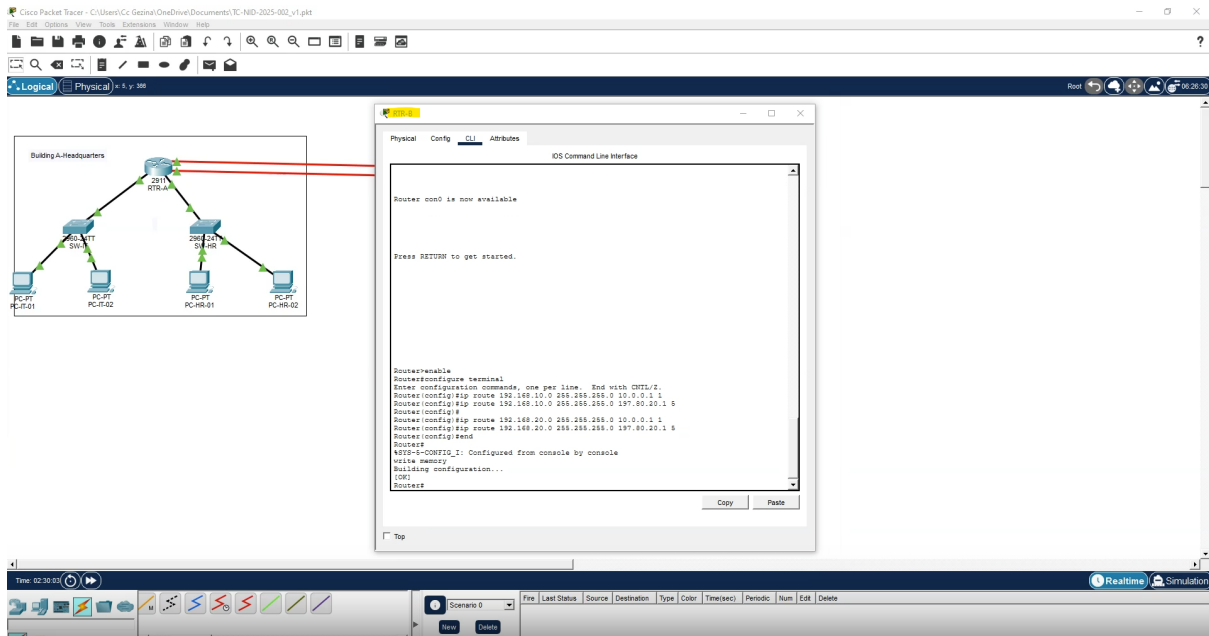


Configured a primary static route (AD 1) via the fibre link and a floating backup route (AD 5) via the ADSL link toward the Finance network.

Why it matters: A five-point gap between the two Administrative Distances leaves headroom to insert additional backup paths later without renumbering existing routes.

FIGURE 5.2

Mirrored Routes on RTR-B

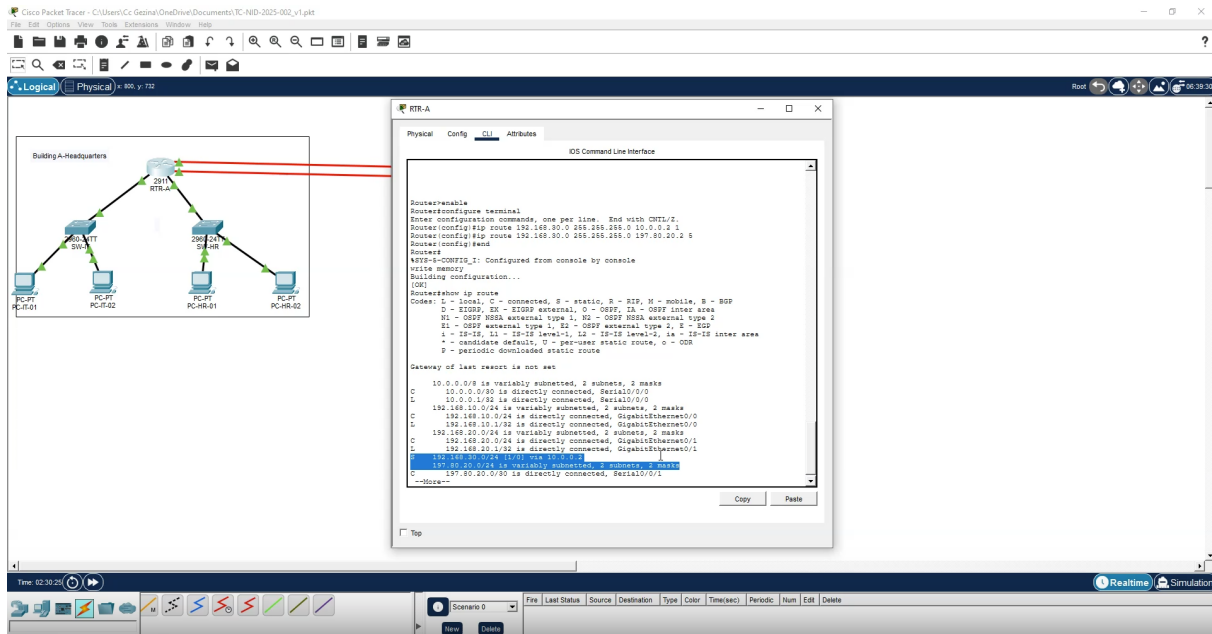


Applied the equivalent primary and backup static routes on RTR-B, pointing back toward the IT and HR networks in Building A.

Why it matters: Mirroring the route policy on both routers ensures failover behaves symmetrically in either direction of traffic, not just from RTR-A outward.

FIGURE 5.3

Routing Table Confirms Floating Behaviour



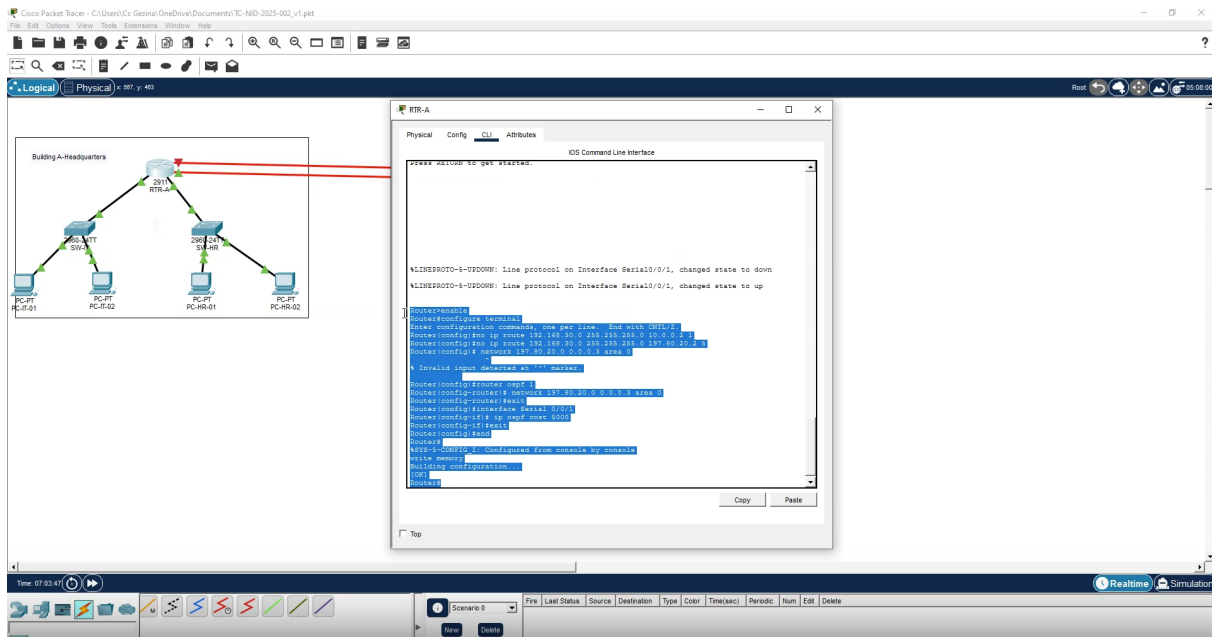
Ran show ip route on RTR-A and confirmed only the AD 1 static route appears; the AD 5 backup route is present in configuration but not installed while the primary path is healthy.

Why it matters: This confirms the floating route is doing exactly what it should — sitting invisible in the background rather than competing with the primary route for use.

As documented in Section 02, the failover-intelligence mechanism was implemented using OSPF rather than IP SLA tracking, due to a Packet Tracer platform limitation. Both WAN interfaces were advertised into the OSPF process on RTR-A and RTR-B, with cost metrics ensuring the fibre link is preferred while the ADSL link remains available as an automatic fallback path.

FIGURE 6.1

OSPF Extended to Both WAN Links on RTR-A

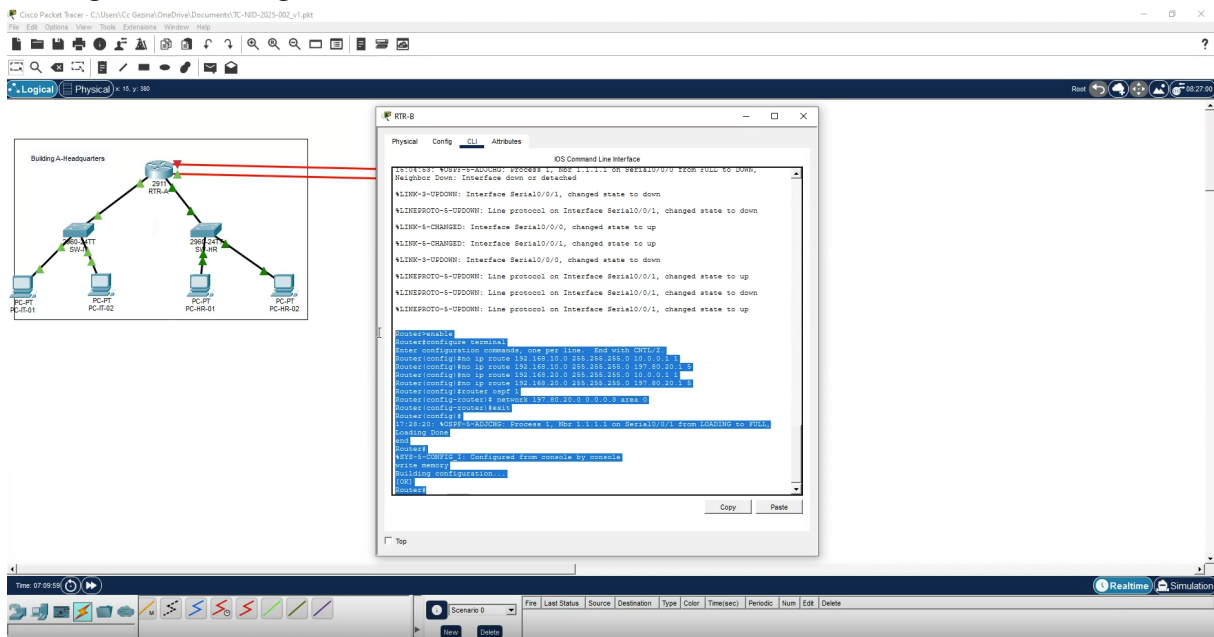


Advertised both Se0/0/0 and Se0/0/1 into OSPF Process 1 Area 0 on RTR-A, allowing the router to dynamically learn and compare paths across both WAN circuits.

Why it matters: Advertising both links into the same OSPF process, rather than only the primary, is what allows OSPF to fail over automatically instead of simply ignoring the backup path.

FIGURE 6.2

Matching OSPF Configuration on RTR-B



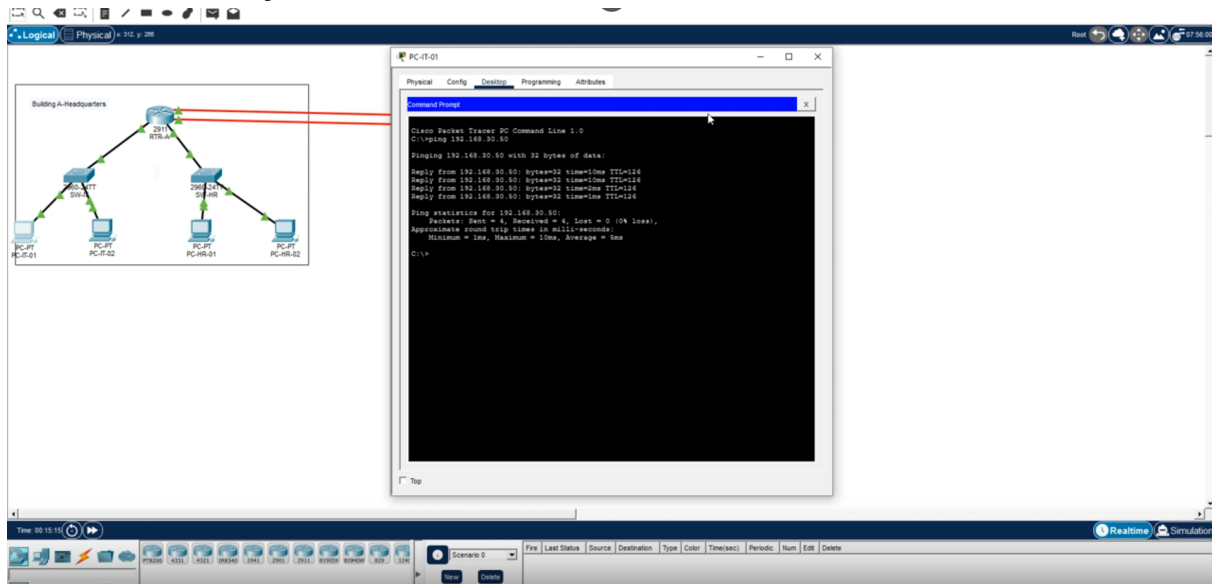
Applied the equivalent OSPF configuration on RTR-B so both routers see two paths to every advertised network and agree on which is preferred.

Why it matters: Consistent OSPF configuration on both ends prevents asymmetric routing, where traffic might leave over one path but return over the other.

The design was validated through controlled failure testing: baseline connectivity was confirmed, the primary fibre link was deliberately removed, traffic was shown to reroute automatically over the ADSL backup with connectivity maintained, and the primary link was then restored, confirming the network automatically reverted to its preferred path with no manual reconfiguration required.

FIGURE 7.1

Baseline Connectivity Confirmed

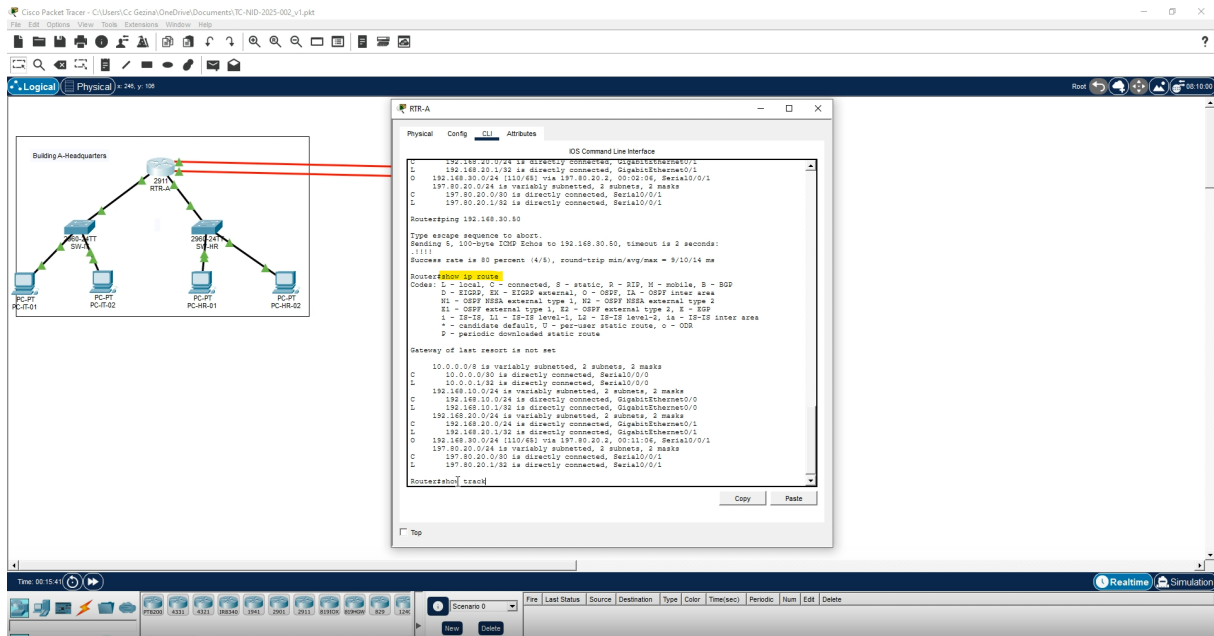


Pinged the Finance server from an IT workstation over the primary fibre link and confirmed successful replies before introducing any fault.

Why it matters: Establishing a clean baseline is what makes the later failover test meaningful — without it there would be no way to distinguish a genuine reroute from a network that was already broken.

FIGURE 7.2

Pre-Failure Routing Table Reviewed

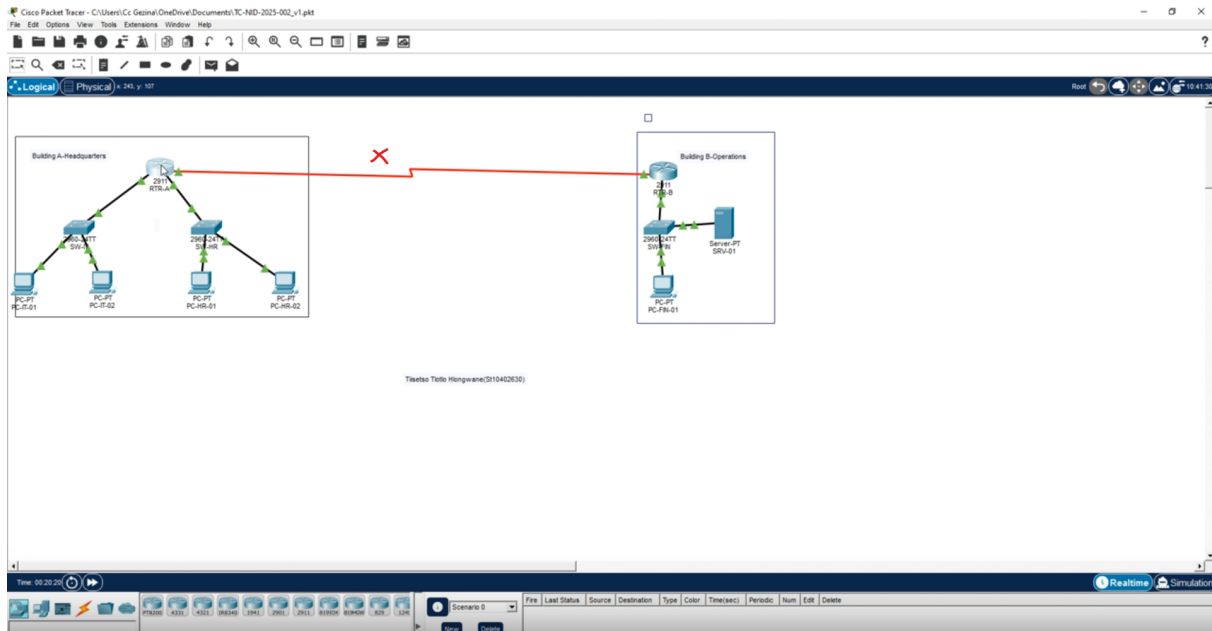


Reviewed show ip route on RTR-A prior to the test, confirming the primary path is installed and the backup path is not yet in use.

Why it matters: Capturing the routing table immediately before the fault gives a direct before/after comparison once the link is broken.

FIGURE 7.3

Primary Fibre Link Deliberately Removed

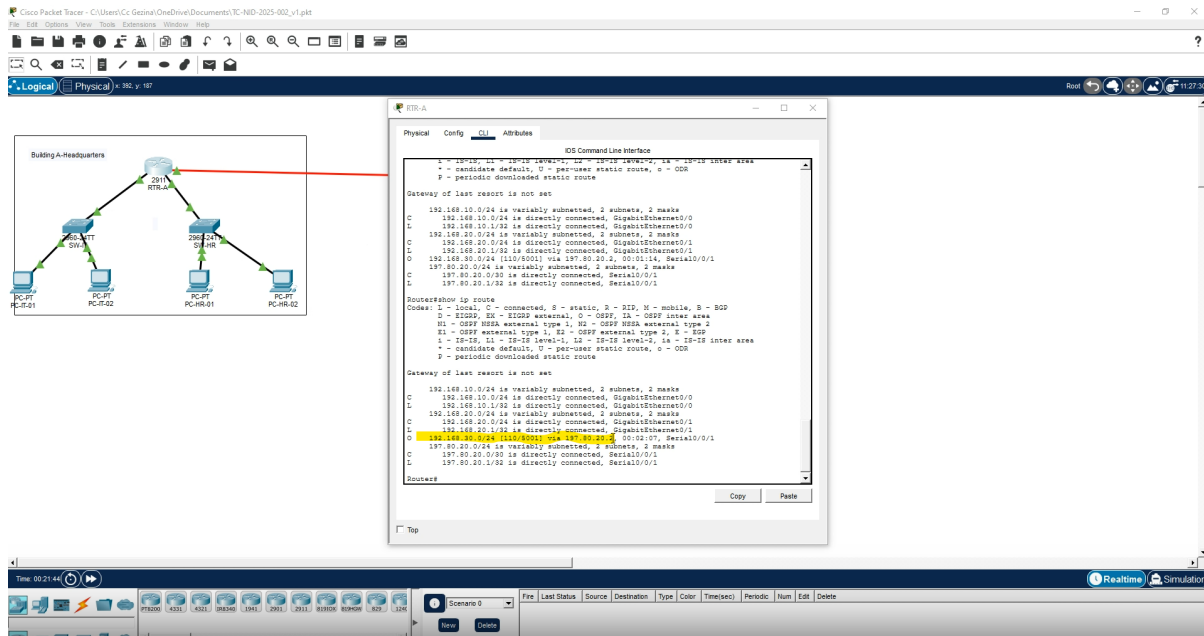


Deleted the cable connecting RTR-A Se0/0/0 to RTR-B Se0/0/0 to simulate a real fibre outage.

Why it matters: Simulating the failure at the physical layer, rather than just shutting the interface down in software, more closely mirrors how a real fibre cut would present to the routers.

FIGURE 7.4

Automatic Reroute Confirmed

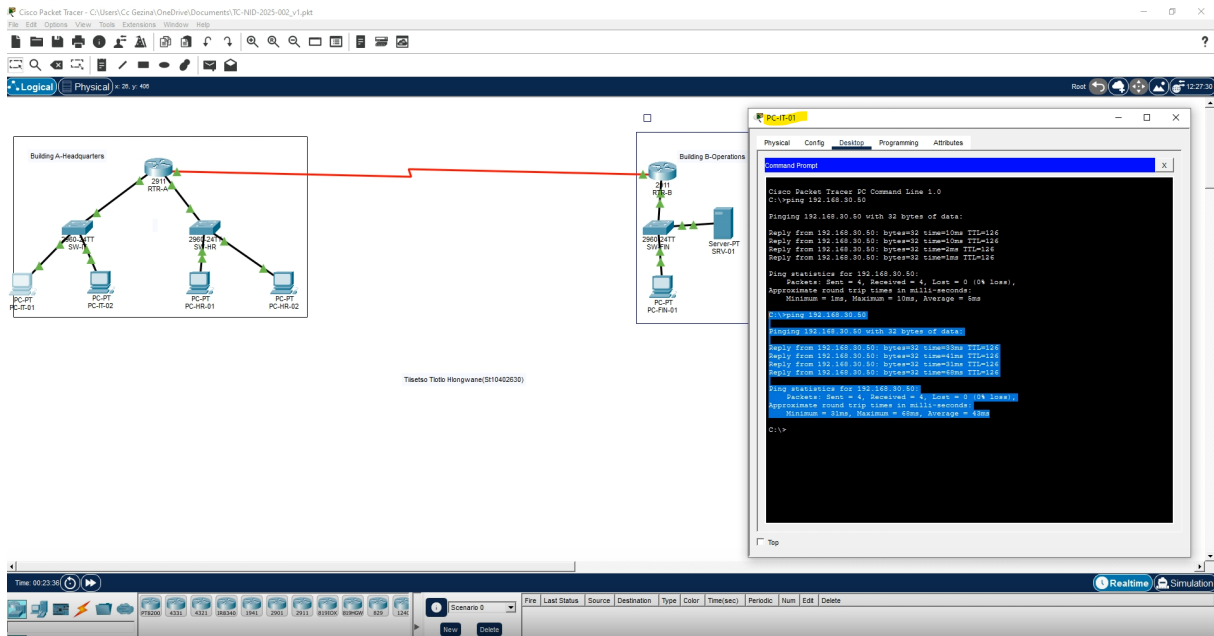


Re-ran the routing and OSPF neighbour checks on RTR-A and confirmed traffic to the Finance network was now flowing over the ADSL backup path within the expected convergence window.

Why it matters: This is the core proof point of the entire project — that failover happens automatically, without any command being typed by an administrator during the outage.

FIGURE 7.5

End-User Connectivity Retested

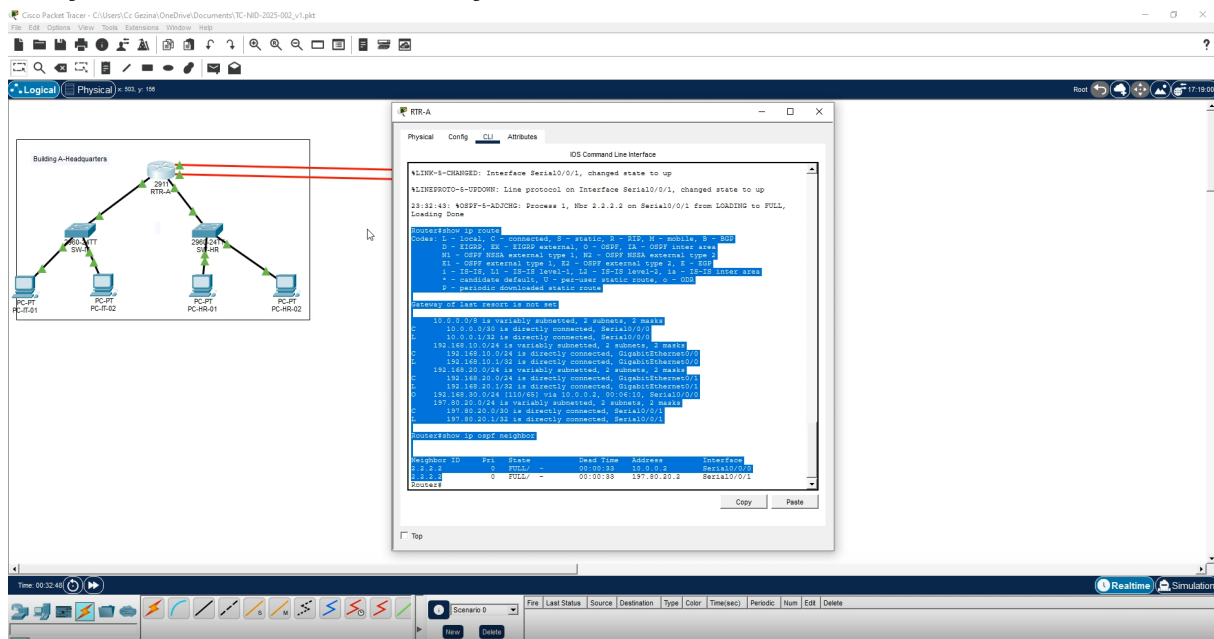


Repeated the ping from the IT workstation to the Finance server and confirmed replies continued successfully, now travelling over the backup ADSL path instead of the fibre link.

Why it matters: Testing from the end-user's perspective, not just the router CLI, confirms the failover is genuinely transparent to the people actually using the network.

FIGURE 7.6

Primary Link Restored and Recovery Verified



Reconnected the primary fibre cable and confirmed the routing table and OSPF adjacency reverted automatically to the preferred fibre path once it came back up.

Why it matters: Validating recovery is as important as validating failure — a design that fails over but never reverts would leave the network permanently on its backup path after every outage.

Outcomes Achieved

Requirement	Result
Redundant physical WAN paths	✓ Independent fibre and ADSL circuits provisioned and addressed
Preferred-path routing	✓ Fibre preferred under normal conditions via lower OSPF cost
Automatic failure detection	✓ OSPF adjacency loss triggers automatic route withdrawal
Automatic failover	✓ Backup ADSL path installed within the OSPF dead interval
Automatic recovery	✓ Primary path reinstated automatically once restored

Technical Skills

Cisco IOS CLI · WAN Redundancy · Floating Static Routes · OSPF Routing · Administrative Distance · Failover Testing · Network Recovery Validation

Competency Assessment

Foundation Level — Enterprise WAN Redundancy & Resilience. Demonstrated ability to extend an existing production topology with a second WAN path, apply Administrative-Distance-based route preference, adapt a failover design around a platform constraint while preserving the original intent, and validate resilience through deliberate, controlled failure and recovery testing rather than assuming correctness from configuration alone.

This portfolio documents the design, build, and validation of a resilient dual-WAN failover architecture for TechCorp, extending the Phase 1 network with a second, independently addressed backup path and an automated mechanism to route around a primary link failure.

The work demonstrates the same administrative discipline carried through every layer: a backup path built as a genuinely independent circuit rather than a shared point of failure; route preference expressed cleanly through Administrative Distance; a failover mechanism that was adapted, not abandoned, when the original plan met a platform limitation; and a resilience claim that was not just configured but proven — tested against a deliberate failure and a full recovery, with routing table and end-user evidence at every stage.

Taken together, this portfolio evidences foundation-level competency in enterprise WAN redundancy design, with a demonstrated ability to translate a resilience requirement into a working, verified, and defensible technical configuration.